

Data Source Documentation

CRE Flood Graph Dashboard — Data Reliability & Provenance

Document Version: 1.0

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Data File: `PowerBI_Dashboard_Package/data/combined_pipeline.csv`

1. Data Sources Overview

The dashboard is powered by a single consolidated CSV file (`combined_pipeline.csv`) containing **999 records** across **2 cities**. The data originates from fundamentally different sources with very different reliability levels.

City	Records	Source	Reliability
Washington, DC	719	WDCEP / Open Data DC	✓ Real / Verified
Atlanta	280	Programmatically generated	⚠ Synthetic / Demonstration Only

2. Washington, DC Data — REAL DATA

Source

- **Primary Source:** Washington DC Economic Partnership (WDCEP) Development Pipeline Reports
- **Portal:** Open Data DC (opendata.dc.gov)
- **Report Years in Dataset:** 2019, 2021, 2025

What It Represents

The DC data tracks **commercial real estate development projects** across the District of Columbia, including:

- Office buildings, retail spaces, multifamily housing, hotels, and institutional projects (“Other”)
- Projects at all stages: Completed, Under Construction, and Pipeline (planned/proposed)
- Data fields include real project names, real street addresses, actual developers, GPS coordinates, estimated square footage, unit counts, and estimated project values

Evidence of Authenticity

- **Real project names** — e.g., “Cybersecurity & Infrastructure Security Agency HQ,” “Salamander Washington D.C.,” “Capital One Arena Modernization,” “Cedar Hill Regional Medical Center”

- **Real developers** — e.g., Georgetown University, Brookfield Properties, Skanska, Tishman Speyer, General Services Administration
- **Real DC addresses** — specific street addresses in NW/NE/SE/SW Washington, DC
- **DC Ward system** — Wards 1-8, matching the actual DC government ward structure
- **Accurate GPS coordinates** — latitude/longitude values that correctly map to Washington, DC (~38.9°N, ~77.0°W)
- **Consistent with publicly verifiable records** — projects like the Federal Reserve Renovation, Smithsonian National Air and Space Museum, and USDA South Building are publicly documented

DC Data by Report Year

Report Year	Records	Notes
2019	425	Historical pipeline snapshot
2021	149	Mid-cycle update
2025	145	Most current report

DC Data by Status

Status	Count
Completed	236
Under Construction	242
Pipeline	241

3. Atlanta Data — SYNTHETIC / DEMONSTRATION DATA

CRITICAL: This Data Is NOT Real

The Atlanta dataset is **programmatically generated sample data** created for demonstration and prototype purposes. It should **NOT** be used for any real analysis, presentations to stakeholders, or decision-making.

Evidence of Synthetic Nature

1. **Coded project names** — All Atlanta projects follow the pattern `ATL-{Sector}-{Number}` :
 - `ATL-Ind-001` , `ATL-Hot-002` , `ATL-Ret-003` , `ATL-Off-013` , `ATL-Mul-007` , `ATL-Mix-009` , `ATL-0th-012`
 - No real project names whatsoever
2. **No architect data** — Every Atlanta record has `NaN` for the architect field (DC records also mostly lack architects, but for different reasons)
3. **Logically impossible status assignments** — Atlanta shows **“Completed” projects in 2026-2027**, which is impossible since those years are in the future:
 - 12 Atlanta projects marked “Completed” with `delivery_year` 2026

- 7 Atlanta projects marked “Completed” with delivery_year 2027
 - This confirms statuses were randomly assigned, not reflecting reality
4. **Evenly distributed data** — The Atlanta data shows suspiciously uniform distributions across sectors and statuses, unlike organic real-world data:
 - Completed: 86, Under Construction: 99, Pipeline: 95 (nearly equal thirds)
 5. **Neighborhood names are real but generic** — Uses real Atlanta neighborhoods (Virginia-Highland, Inman Park, Beltline Westside, Old Fourth Ward, etc.) but they appear randomly assigned to fake projects
 6. **Developer names are real but recycled** — Uses names of real Atlanta developers (Holder Properties, New City Properties, Carter, Selig Enterprises) but assigned randomly
 7. **Report years span unnaturally evenly** across 2018-2025

Why Synthetic Data Exists

The Atlanta data was likely generated to:

- Demonstrate the dashboard’s multi-city comparison capabilities
- Show how the tool would work with data from additional markets
- Provide a prototype for what a full production version could look like

4. Time Periods — Understanding 2026-2030 Data

This is a critical area of concern. Here’s the breakdown:

DC Data (2026-2030) — REAL Pipeline/Planned Projects

For Washington, DC, projects with delivery years 2026-2030 represent **real planned projects** that are currently in the development pipeline or under construction. These are NOT forecasts or projections — they are actual projects with approved plans, identified developers, and estimated delivery timelines.

Year	DC Records	Status Breakdown
2026	12	0 Completed, 14 Under Construction, 0 Pipeline
2027	16	0 Completed, 3 Under Construction, 25 Pipeline
2028	6	0 Completed, 0 Under Construction, 6 Pipeline
2029	3	0 Completed, 2 Under Construction, 1 Pipeline
2030	5	0 Completed, 0 Under Construction, 5 Pipeline
2033	1	0 Completed, 0 Under Construction, 1 Pipeline

Note: DC’s future-year data is logically consistent — no DC projects are marked “Completed” for years that haven’t occurred yet. Projects in 2026+ are all either “Under Construction” or “Pipeline,” which is exactly what you’d expect.

Atlanta Data (2026-2027) — SYNTHETIC ⚠️

Atlanta’s future-year data has **impossible combinations** (e.g., “Completed” in 2026), confirming it’s synthetic:

Year	Atlanta Records	Includes “Completed”
2026	43	Yes (12 records) — impossible
2027	25	Yes (7 records) — impossible

160 Records with Unknown Delivery Year

- All 160 records with missing delivery years are DC projects
 - These are Pipeline projects where no delivery timeline has been established
 - This is normal for early-stage pipeline projects
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5. Data Reliability Matrix

Data Segment	Reliability Rating	Use For	Do NOT Use For
DC — Completed (2024 and earlier)	● Verified / Real	Analysis, presentations, reports	—
DC — Under Construction (2025-2029)	● Real / Current	Pipeline analysis, market outlook	Guarantees of completion
DC — Pipeline (2025-2033)	● Real but Tentative	Market intelligence, trend analysis	Firm delivery commitments
DC — No delivery year (Pipeline)	● Real but Early-Stage	Awareness of planned projects	Timeline-dependent analysis
DC — Historical (2019 report data)	● Verified Historical	Trend analysis, historical context	Current market state
Atlanta — All data	● Synthetic / Demo	Dashboard prototype demos	Any real analysis or presentations

6. Limitations & Caveats

Known Limitations

1. **Atlanta data is entirely synthetic** — Cannot be used for any real-world conclusions about the Atlanta CRE market.
2. **DC Pipeline projects are subject to change** — Pipeline status means the project is proposed/planned but could be delayed, redesigned, or cancelled. Delivery year estimates for Pipeline projects are especially tentative.
3. **Historical snapshots, not continuous tracking** — The DC data comes from periodic WDCEP reports (2019, 2021, 2025), not continuous monitoring. Some projects may appear in multiple report years.
4. **Duplicate records exist** — Some DC projects appear multiple times (e.g., across different sectors for mixed-use projects, or across report years). This inflates project counts if not deduplicated.
5. **Estimated values are inconsistent** — Many records have \$0 or missing estimated values, making total value calculations unreliable.
6. **No real-time updates** — The dataset is a static snapshot and does not auto-update as projects progress.
7. **Sector classification** — Some projects span multiple sectors. The “Hotel” sector includes cultural venues (Smithsonian, Ford’s Theatre) alongside actual hotels, as WDCEP classifies by primary use or development context.

7. Recommendations

For Prototype / Demo Use

- The dashboard works well as a **prototype** showing how a multi-city CRE pipeline tracker would function
- DC data is suitable for demonstrating real analytical capabilities
- Atlanta data adequately shows multi-city comparison features

For Production Use — Required Steps

1. **Replace Atlanta synthetic data** with real data from:
 - Atlanta Regional Commission (ARC)
 - Invest Atlanta
 - CoStar Group / CBRE / JLL market reports
 - Atlanta Business Chronicle development tracker
2. **Deduplicate DC records** — Cross-reference across report years to avoid double-counting
3. **Add data freshness indicators** — Show when each record was last updated
4. **Implement automated data refresh** — Connect to WDCEP Open Data DC API for periodic updates
5. **Add data quality flags** — Mark records with missing values, tentative timelines, or early-stage pipeline status
6. **Label synthetic data clearly** — If keeping Atlanta demo data, add a visible “SAMPLE DATA” watermark or banner

For Presentations

-  DC data can be presented as real market intelligence
-  Atlanta data should be clearly labeled as “illustrative/demonstration” in any presentation
-  Future-year DC Pipeline projects should be presented as “planned/proposed” not “confirmed”